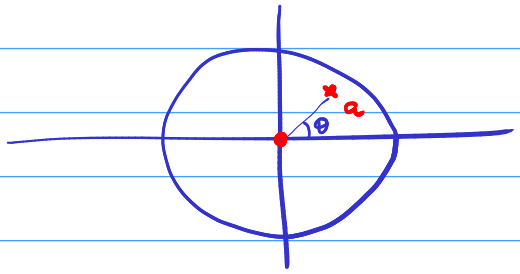
Magnitude Response

First order system — Single pole

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1 - az^{-1}} \quad ; \quad |z| > |a|$$

$$(1 - az^{-1})Y(z) = X(z)$$

$$h(n) = a^n u(n)$$



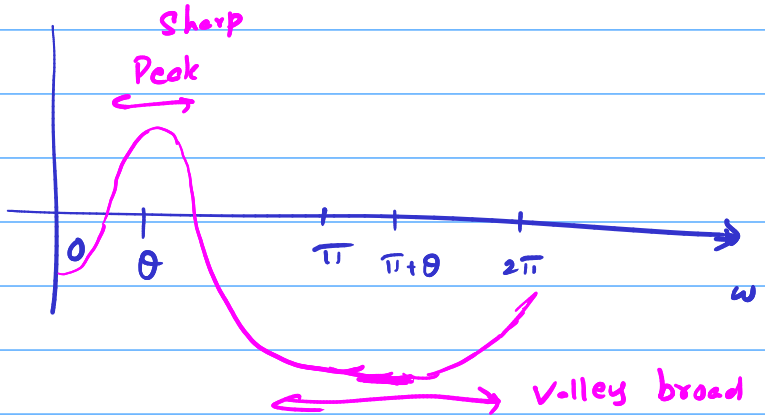
$$y(n) = x(n) + a\,y(n-1)$$

$|a| < 1$ for stability

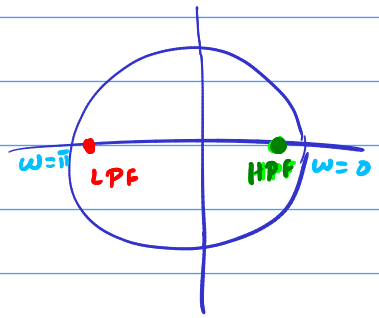
$$H(\omega) = \frac{1}{1 - ae^{-j\omega}} \qquad a = re^{j\theta} \qquad r < 1$$

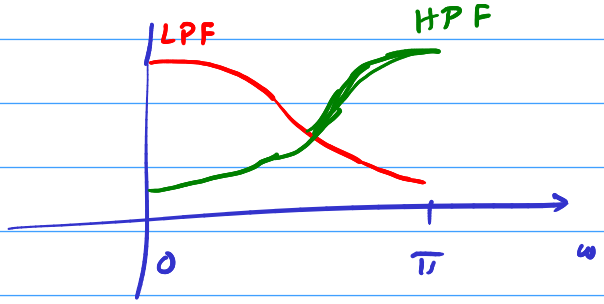
$20 \log |H(\omega)|$ = − (ve) of response of a zero at $re^{j\theta}$

$$= -20 \log (1 + r^2 - 2r \cos(\omega - \theta))$$

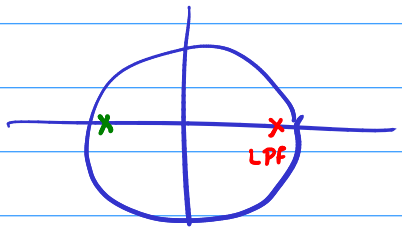


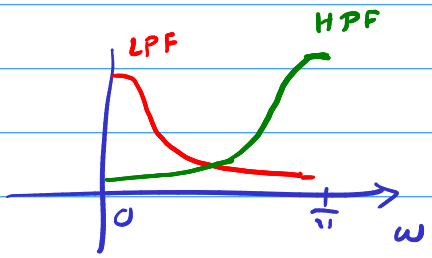
Low Pass & High pass filters using zeros





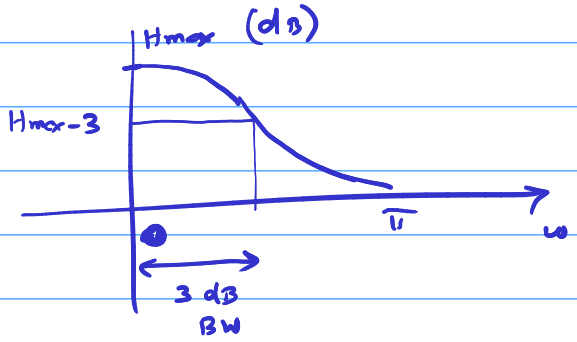
LPF & HPF using poles





Definition: **3 dB Bandwidth**

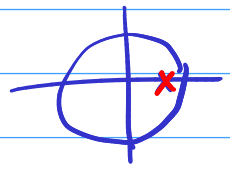
The size of band for which the magnitude gain is within 3 dB of max. gain



3 dB BW for

Singe pole

$$H(z) = \frac{1}{1 - 0.9 z^{-1}}$$



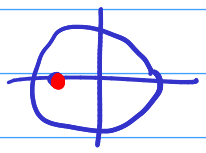
$\omega = 0 \quad |H(\omega)|^2 = \frac{1}{(1-0.9)^2} = 100$

$|H(\omega = \pi/30)|^2 \approx 50$

3 dB BW $\approx \pi/30$

Single zero Syst

$$H(z) = 1 + 0.9 z^{-1}$$



$\omega = 0 \quad |H(\omega)|^2 = (1 + 0.9)^2$

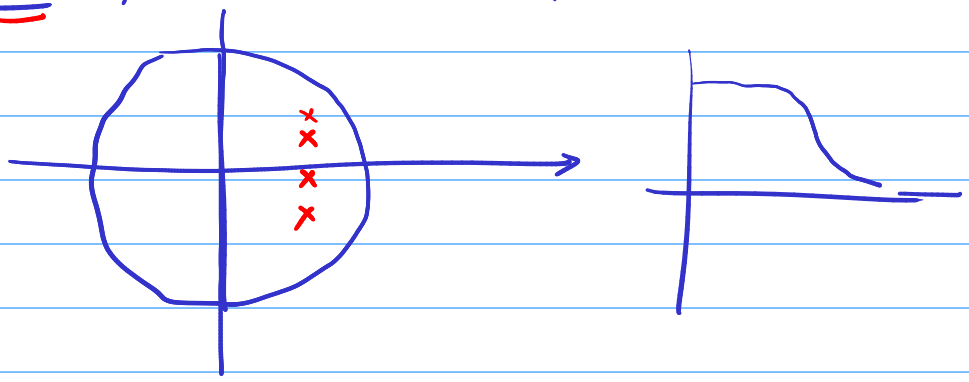
$= 1.9^2 = 3.6$

$\omega = \pi/2 \quad |H(\omega = \pi/2)|^2 \approx 1.8$

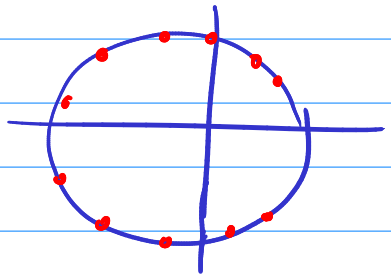
3 dB BW $\approx \pi/2$



For Flat response we add more poles



Using Zeros we get Similar response

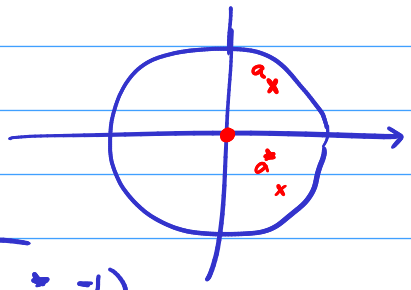


order of system will be higher with zero.

Second order System

Resonator

Poles at a, a^*



$$H(z) = \frac{1}{(1-az^{-1})} \cdot \frac{1}{(1-a^*z^{-1})}$$

$$a = re^{j\theta}$$

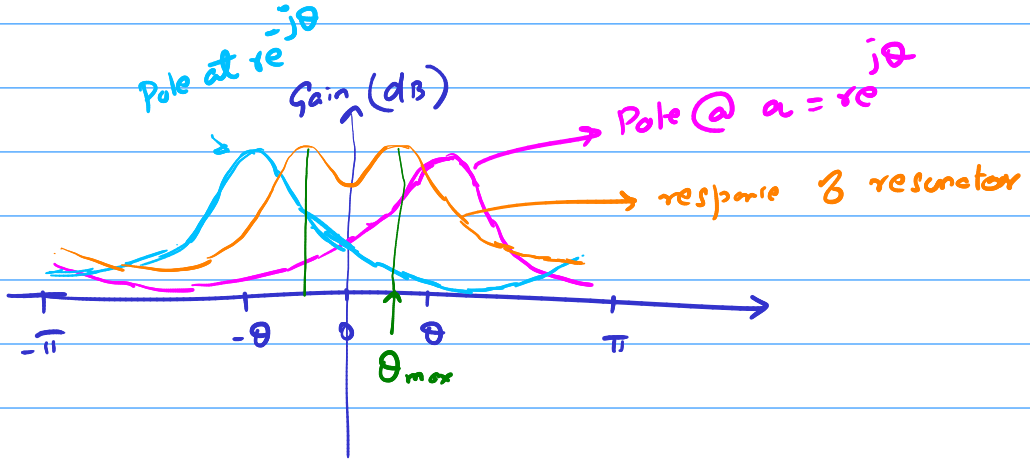
$$a^* = re^{-j\theta}$$

$$h(n) = r^n \frac{\sin(n+1)\theta}{\sin\theta} u(n)$$

$$|H(\omega)| = \frac{1}{|1-ae^{-j\omega}|} \cdot \frac{1}{|1-a^*e^{-j\omega}|}$$

$$20\log|H(\omega)| = -20\log|1-ae^{-j\omega}| \Rightarrow \text{Pole @ } a$$

$$-20\log|1-a^*e^{-j\omega}| \Rightarrow \text{pole @ } a^*$$

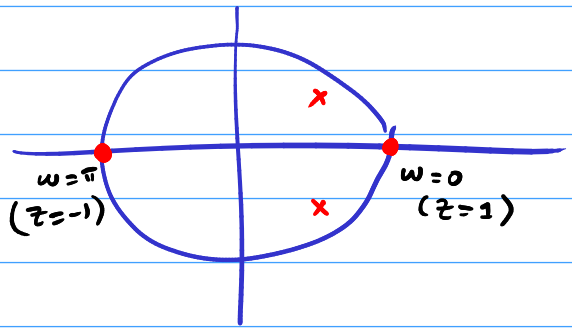


$$\theta_{max} = \cos^{-1}\left(\frac{1+r^2}{2r} \cdot \cos\theta\right)$$

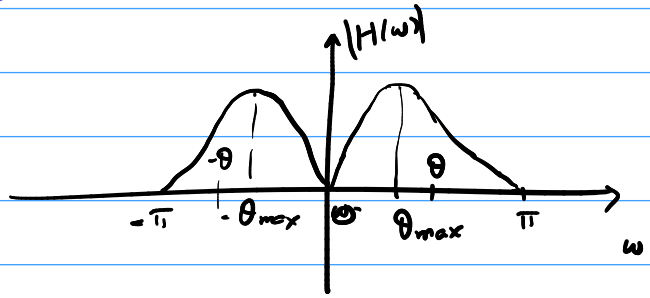
As $r \to 1$, $\theta_{max} \to \theta$.

If $\theta = \pi/2$ then $\theta_{max} = \pi/2$

Band Pass Filter using resonator



$$H(z) = \frac{(1 - z^{-1})(1 + z^{-1})}{(1 - a z^{-1})(1 - a^{*} z^{-1})}$$



$a = re^{j\theta}$

$$\theta_{max} = \cos^{-1}\left(\frac{2r}{1 + r^2}\cos\theta\right)$$